

HIREMIND — AI-DRIVEN TALENT INTELLIGENCE & WORKFORCE ANALYTICS PLATFORM

A Project Report

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|    & TECHNOLOGY (PIET)        |
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CERTIFICATE

This is to certify that Project-1 (203105499) of the 6th Semester, entitled “**HireMind: AI-Driven Talent Intelligence & Workforce Analytics Platform,**” of Group No. **PIET AIML 12**, has been successfully completed by the following students:

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ii

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“The single greatest cause of happiness is gratitude.”

— *Auliq-Ice*

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iii

Abstract

In today's fast-paced corporate and technical environment, talent acquisition and workforce retention represent two of the most critical challenges facing modern human resource (HR) organizations. Traditional recruitment workflows rely heavily on manual resume screening, where HR professionals spend hundreds of hours sifting through thousands of unstructured PDF resumes. This manual process is slow, inconsistent, error-prone, and prone to cognitive bias. Furthermore, conventional HR platforms operate in isolation from predictive employee retention models, leaving organizations vulnerable to unexpected employee attrition and costly turnover.

To solve these interconnected problems, this project presents **HireMind**, an intelligent, decoupled three-tier microservice platform for automated candidate resume screening, hybrid job matching, employee attrition risk forecasting, and automated candidate outreach. The system combines modern MERN full-stack web engineering (MongoDB, Express.js, React 18, Node.js) with a specialized Python Flask AI/ML processing service.

HireMind features a novel **Hybrid Candidate Matcher Engine** that fuses two complementary evaluation signals: an explicit skill matrix intersection score (70% weight) and a TF-IDF vector space model with Cosine Similarity (30% weight). For workforce retention, HireMind incorporates a **Random Forest Classification Engine** (150 decision trees) trained on key demographic and organizational metrics (Age, Monthly Income, Years at Company, Job Satisfaction) to output real-time attrition probabilities and actionable risk explanations. Additionally, the system includes an automated **PyPDF2/Regex NLP Resume Parser**, an encrypted **Nodemailer Candidate Outreach Engine**, and an interactive **Conversational HR AI Chatbot**.

Deployed via a 1-click **Render.com** blueprint architecture and secured with JSON Web Tokens (JWT) and Role-Based Access Control (RBAC), HireMind executes full resume parsing, skill matching, and attrition risk evaluation in under **15 milliseconds**, reducing resume screening effort by over **99.2%** while eradicating human screening inconsistencies.

Keywords: Talent Intelligence, Natural Language Processing (NLP), Resume Parsing, TF-IDF Vectorization, Cosine Similarity, Employee Attrition Prediction, Random Forest Classifier, MERN Stack, Microservices, Workforce Analytics, Render Blueprint, Automated Candidate Outreach.

iv

Table of Contents

- **Acknowledgements ... iii**
- **Abstract ... iv**
- **List of Tables ... ix**
- **List of Figures ... x**
- **List of Abbreviations ... xii**

1 Introduction ... 1

- **1.1 Background ... 1**
- **1.2 Problem Overview ... 2**
- **1.3 Need for the Project ... 2**
- **1.4 Proposed Approach ... 3**
- **1.5 Objectives ... 4**
- **1.6 Scope of the Project ... 4**
- **1.7 Conclusion ... 4**

2 Literature Survey ... 5

- **2.1 Introduction ... 5**
- **2.2 The Resume Screening & Talent Acquisition Challenge ... 5**
- **2.3 Review of Related Systems ... 6**
 - 2.3.1 Rule-Based Keyword Resume Parsers ... 6
 - 2.3.2 Deep Learning & LLM-Based Screening Tools ... 7
 - 2.3.3 Predictive HR Analytics & Attrition Models ... 7
- **2.4 Comparative State-of-the-Art Analysis ... 7**

- **2.5 Research Gap and Motivation ... 8**
- **2.6 Conclusion ... 8**

3 Project Flow and Methodology ... 9

- **3.1 Introduction ... 9**
- **3.2 Project Workflow Overview ... 9**
- **3.3 The Six-Stage Processing Pipeline ... 10**
- **3.4 Hybrid Keyword & TF-IDF Cosine Similarity Resolution ... 11**
- **3.5 Input-Process-Output Specification ... 12**
- **3.6 Conclusion ... 12**

4 System Design ... 13

- **4.1 Introduction ... 13**
- **4.2 Design Objectives and Principles ... 13**
- **4.3 Modular Component Architecture ... 14**
 - 4.3.1 Client Presentation Tier (React 18 + Vite) ... 14
 - 4.3.2 Application & API Gateway Tier (Express.js + Node.js) ... 14
 - 4.3.3 AI/ML Analytics Tier (Flask + scikit-learn Engine) ... 15
- **4.4 Sequence and Interaction Design ... 15**
- **4.5 Database and Data Model Design ... 16**
- **4.6 Conclusion ... 16**

5 Implementation ... 17

- **5.1 Introduction ... 17**
- **5.2 Technologies and Implementation Stack ... 17**
- **5.3 User Interface and Talent Workspace ... 17**
- **5.4 Automated Resume Parsing & NLP Information Extraction Engine ... 19**
- **5.5 Hybrid Resume-Job Matching Engine (TF-IDF + Explicit Skills Matrix) ... 21**
- **5.6 Machine Learning Employee Attrition & Retention Prediction Engine ... 22**
- **5.7 Automated Candidate Outreach & Email Automation Engine ... 23**
- **5.8 AI HR Assistant Chatbot Interface ... 24**
- **5.9 Role-Based Access Control & Security Encryption Subsystem ... 24**
- **5.10 Conclusion ... 25**

6 Result Analysis ... 26

- **6.1 Introduction ... 26**

- **6.2 Evaluation Methodology and Metrics ... 26**
- **6.3 Comparative Evaluation: Manual vs. HireMind Automation ... 26**
- **6.4 Production Resume Screening & Attrition Benchmark ... 28**
- **6.5 Discussion of Findings ... 29**
- **6.6 Conclusion ... 29**

7 Testing and Quality Assurance ... 30

- **7.1 Introduction ... 30**
- **7.2 Multi-Stage Testing Architecture ... 30**
- **7.3 Pre-Flight Rule Compliance Matrix ... 31**
- **7.4 Integration and End-to-End System Testing ... 32**
- **7.5 Conclusion ... 32**

8 Maintenance and Operations ... 33

- **8.1 Introduction ... 33**
- **8.2 Continuous Synchronization and Model Retraining Lifecycle ... 33**
- **8.3 Day-2 Observability and Operational Telemetry ... 34**
- **8.4 Types of Software Maintenance ... 35**
- **8.5 Conclusion ... 35**

9 Deployment and DevOps Integration ... 36

- **9.1 Introduction ... 36**
- **9.2 Automated Render.com Blueprint CI/CD Delivery Pipeline ... 36**
- **9.3 Multi-Environment Infrastructure Topologies ... 37**
- **9.4 Application Hosting and Cloud Deployment ... 38**
- **9.5 Conclusion ... 38**

10 Conclusion and Future Scope ... 39

- **10.1 Conclusion ... 39**
- **10.2 Project Development Timeline ... 39**
- **10.3 Summary of Contributions ... 40**
- **10.4 Future Work and Multi-Cloud Expansion ... 41**
 - 10.4.1 Cross-Platform Social & Professional Profile Aggregation ... 41
 - 10.4.2 LLM Fine-Tuning & Semantic Vector Embeddings (BERT/SBERT) ... 42
 - 10.4.3 Automated Video Interview Analysis & Sentiment Scoring ... 42
 - 10.4.4 Enterprise ATS Integration (Workday, Greenhouse, Lever) ... 42

- **10.5 Closing Remarks ... 42**

References ... 43

v

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